

A 16-Bit-Native Compression Pipeline for Lossless Medical Images

(Plain-Language Edition)

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Abstract—Medical images must be stored and transmitted losslessly, since a radiologist must see exactly what the scanner produced, and they are “deep,” carrying 10–16 bits per pixel rather than the 8 bits of an ordinary photo. That deep format is where most compression tools perform poorly, because nearly all of them were built for 8-bit data. This paper describes MIC (Medical Image Codec), a lossless codec that works directly on 16-bit data instead of splitting each pixel into two bytes. MIC chains three simple stages, a spatial predictor, a 16-bit run-length encoder, and a large-alphabet entropy coder called Finite State Entropy (FSE, a fast table-driven form of asymmetric numeral systems), and adds a multi-state decoder that keeps the CPU more fully utilized to speed up decompression. On 21 real DICOM images across nine modalities, MIC compresses 5–22% better (14% on average) than a strong general-purpose baseline (Delta + Zstandard), reaches an average compression ratio of 3.46× (essentially tied with High-Throughput JPEG 2000 at 3.45× and about 91% of JPEG-LS at 3.82×), and is the fastest decoder of the four tested codecs on 20 of 21 images on ARM64 and all 21 on AMD64. A 20 KB pure-JavaScript decoder lets the format be decoded directly in a web browser. The main caveat is that this is a 21-image study; the results are encouraging but need confirmation on larger, multi-hospital datasets.

Index Terms—medical image compression, lossless compression, entropy coding, asymmetric numeral systems (ANS), finite state entropy (FSE), DICOM, high-bit-depth imaging.

I. WHY MEDICAL IMAGES NEED A DIFFERENT CODEC

A MODERN scanner produces a great deal of data. A single digital breast tomosynthesis view can comprise 60 or more image slices and over 600 MB; a full screening study can exceed 2 GB uncompressed. Hospitals store and move very large volumes of this data every day, so effective lossless compression directly affects storage cost, network load, and how quickly images appear in a viewer.

DICOM [1], the medical imaging standard, already supports several lossless formats—JPEG 2000 [2], HTJ2K [3], JPEG-LS [5], and a basic run-length scheme. Each makes a different trade-off between how small the file gets, how fast it decodes, and how complex it is to implement [18], [19]. JPEG 2000 and HTJ2K compress well but rely on fairly heavy machinery. JPEG-LS usually gets the smallest files. Plain run-length coding is simple but barely compresses.

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This is a plain-language edition of the MIC paper. It keeps the main ideas, key numbers, and reasoning, but trades the formal equations and implementation minutiae for explanations a software engineer can follow. Data tables are reproduced from the full manuscript; figures are omitted.

A. The core problem: most coders assume 8-bit data

This is the central observation behind the paper. The widely used entropy coders, including Huffman, arithmetic coding, the LZ77 family, and the FSE/ANS coder inside Zstandard [11], were all designed for **8-bit byte streams**, where there are only 256 possible symbols. Medical images do not fit that assumption: a 16-bit pixel has up to 65,536 possible values.

When an 8-bit coder is applied to 16-bit data, it has two poor options:

- 1) **Split each pixel into a high byte and a low byte** and code them separately. This doubles the number of coding steps and, more importantly, discards the relationship between the two bytes.
- 2) **Cap the alphabet at 256 values** and store anything rarer as raw data, which wastes space.

Why does splitting hurt? Consider a tiny prediction error like -2 , -1 , 0 , or $+1$. The low byte carries the real information, but the high byte is almost always the same (just a sign-extension). A byte-by-byte coder still spends bits encoding that high byte even though it is nearly fully determined by the low one. On the test images, the information shared between the two bytes ranges from 0.3 to 1.1 bits per pixel, which is the ceiling on how much a byte-splitting coder is wasting, and MIC’s measured 5–22% advantage falls within that ceiling.

a) *The residual stream:* Throughout, a *residual* is the difference between a pixel and a prediction of it. Example: if the predictor guesses 1020 and the true pixel is 1023, the residual is $+3$. Smooth regions of an image produce many residuals near zero, and the same small values recur frequently, which is exactly the pattern run-length and entropy coding exploit well.

B. What MIC is

MIC keeps the residuals at their native 16-bit width and combines (i) a simple spatial predictor, (ii) a 16-bit run-length encoder, and (iii) an entropy coder (FSE) extended to handle large alphabets. Plus a multi-state decoder that improves decompression speed by letting the CPU work on several symbols at once.

C. Contributions, briefly

- **A 16-bit-native pipeline** (Delta + 16-bit RLE + extended FSE) that outperforms Delta + Zstandard on every one of the 21 images.
- **An entropy coder for large alphabets**—up to 65,535 symbols, versus the 4,096-symbol limit in Zstandard’s

FSE. Tables shrink automatically to fit the data (as small as ~2 KB for 8-bit content).

- A **multi-state decoder** that runs 2/4/8 independent decode chains to use idle CPU capacity, with no cost to compression ratio.
- An **evaluation** against HTJ2K, JPEG-LS, and Delta + Zstandard.
- A **20 KB JavaScript decoder** plus a WebAssembly build, for client-side decoding in browsers.

a) *Scope*: This work covers lossless, single-frame, grayscale images only. Color and multi-frame extensions, lossy modes, and progressive decoding are out of scope.

II. RELATED WORK

The DICOM lossless codecs. JPEG 2000 [2] uses a wavelet transform plus a sophisticated block coder (EBCOT): strong ratios, but slow decoding. HTJ2K [3], [4] replaces the block coder with a faster one and decodes considerably quicker. JPEG-LS [5] uses a small predictor (Median Edge Detector) and Golomb-Rice coding; it is a widely deployed standard and a strong ratio baseline. The basic DICOM run-length scheme compresses little because it works on raw bytes with no prediction step. Research codecs such as CALIC [6] and FLIF [7] compress well on natural images but are not DICOM standards and lack browser decoders.

These are the codecs MIC is measured against. The goal is not to outperform them on every metric, but to explore a **simpler, residual-domain design** aimed at high-bit-depth images, with strong speed and easy deployment.

Entropy coding background. Asymmetric Numeral Systems (ANS) [8], [9] is a family of entropy coders that replaces arithmetic coding’s interval math with simple integer state updates. FSE is a fast, table-driven version of ANS [10] made popular by Zstandard [11]. The limitation is that these implementations are tuned for byte streams (Zstandard caps FSE at 4,096 symbols).

Parallel ANS. Prior work has shown that ANS decoders run faster when several independent streams are decoded at once (Giesen on interleaved coders [12]; later work scaling this to many streams [13] and to GPUs [14]). MIC does not claim to invent ANS or interleaving; it applies these known ideas inside a 16-bit-native medical codec with a specific large-alphabet design.

Table I places MIC against the prior work across five dimensions relevant to medical image compression.

III. HOW MIC WORKS

The pipeline is three stages, each simple enough to decode in one straight-line pass with very few branches: raw 16-bit pixels go through spatial prediction (subtract a guess based on neighbors), then 16-bit run-length encoding (collapse repeated values), then FSE entropy coding (remove the remaining redundancy), producing compressed bytes.

A. Spatial prediction

For each pixel, MIC predicts its value as the **average of the pixel above and the pixel to the left**, then stores only the

TABLE I

FEATURE COMPARISON OF MIC WITH RELATED CODECS AND ENTROPY CODING METHODS. “ENTROPY CODER SEES 16-BIT RESIDUAL AS ONE SYMBOL” IS A PROPERTY OF THE ENTROPY STAGE, NOT OF THE CODEC’S OVERALL BIT-DEPTH SUPPORT. JPEG-LS AND HTJ2K BOTH HANDLE 16-BIT MEDICAL IMAGES, BUT THEIR ENTROPY STAGES (GOLOMB-RICE ON CONTEXT BINS FOR JPEG-LS, BIT-PLANE CODING FOR HTJ2K) DO NOT TREAT THE PREDICTED RESIDUAL AS A SINGLE 16-BIT SYMBOL. SOURCE-LINE COUNTS ARE ORDER-OF-MAGNITUDE CONTEXT ONLY; THE REFERENCE LIBRARIES ALSO IMPLEMENT STANDARD PROFILES OUTSIDE MIC’S CURRENT SCOPE.

Feature	JPEG-LS	HTJ2K	Zstd	rANS	MIC
Lossless medical	Yes	Yes	General	General	Yes
Entropy coder sees 16-bit residual as one symbol	No	No	No	Configurable	Yes
Multi-state ILP	No	No	No	Yes	Yes
Browser decoder	Via WASM port	Via WASM port	Partial	No	Purpose-built JS
Source lines	5k	20k	N/A	N/A	1.1k

difference (the residual). Pixels on the top edge or left edge use whatever single neighbor is available; the very first pixel is stored as-is. Division rounds down, and the decoder repeats the exact same arithmetic so reconstruction is exact.

This predictor was chosen because it is inexpensive, has almost no branches on the decode side, and works well on medical images.

a) *Other predictors we evaluated*: We compared four predictors (left-only, average, Paeth from PNG [16], and MED from JPEG-LS [5]) through the full pipeline. MED gave the best ratio (about 1.6% better than average) but decoded 1.5–2× slower because of its three-way branch and a diagonal dependency that blocks the fast branch-free loop. The average predictor was preferred for its consistency and speed, so it is the default. Table II summarizes the comparison.

TABLE II

PREDICTOR ABLATION SUMMARY: GEOMETRIC MEANS AND WIN COUNTS ACROSS ALL 21 IMAGES.

Predictor	Geo. Mean	Wins	Avg→X
Left-only	3.38×	3/21	−2.3%
Average (MIC)	3.46×	13/21	baseline
Paeth	3.48×	2/21	+0.5%
MED (JPEG-LS)	3.52×	16/21	+1.6%

B. Handling large jumps: overflow coding

Most residuals are small, but occasionally a pixel jumps far from its prediction. Rather than widen the whole stream to 32 bits to handle rare large values, MIC reserves one special code as a **delimiter**: small, in-range residuals are stored directly as 16-bit codes (with zero mapped to a fixed midpoint value); a large, out-of-range residual is stored as the delimiter followed by the raw pixel value.

On decode, the reader checks each value: if it is the delimiter, the next value is a raw pixel; otherwise it is a normal residual. The ranges are designed not to overlap, so there is never any ambiguity. In normal 8–16-bit clinical data the overflow path is rare; it exists to guarantee correctness, not for the common case. Table III shows the mapping for a representative bit depth.

C. 16-bit run-length encoding (RLE)

The residual stream is turned into two kinds of runs: **same runs** (a count plus one repeated value, used when a value

TABLE III
OVERFLOW CODING EXAMPLE FOR $d = 12$ BITS ($\Delta\text{THRESHOLD} = 2047$,
DELIMITER $D = 4095$).

Residual r	Emitted code(s)	Notes
-2	2045	in-range
-1	2046	in-range
0	2047	in-range (= threshold)
+1	2048	in-range
+2	2049	in-range
± 2047	4095, raw pixel	overflow

appears at least three times in a row) and **literal runs** (a count followed by that many values copied verbatim; the values need not be distinct). A single header number tells the two apart using a midpoint trick: small header values mean “same run of this length,” large ones mean “literal run.” The longest single run is $\sim 32,767$ symbols; longer stretches are split across several headers automatically.

Why 16-bit RLE matters. Suppose 1,000 identical 16-bit residuals appear in a row. MIC encodes that as just two numbers: a count and the value. A *byte*-oriented RLE sees the high and low bytes interleaved, so no single byte repeats 1,000 times, so it is forced into two interleaved runs or falls back to literals. Working at 16 bits captures the repetition that byte-splitting hides.

Fast decode. Same runs (often 80% of all runs on smooth images) are fast-pathed: the decoder just hands back the cached value and decrements a counter, without touching the compressed data at all.

D. The entropy coder (FSE) for large alphabets

FSE in one paragraph. FSE keeps a single integer “state.” To decode one symbol, it looks up the current state in a table; the entry tells it which symbol to output, how many bits to read next, and how to compute the next state. So each symbol costs **one table lookup, one bit read, and one addition**—no multiplies or divides. (Encoding runs the symbols in reverse and builds the bitstream backwards; decoding reads forwards. That reversal is inherent to how ANS works.)

MIC extends FSE to handle up to **65,535 symbols** instead of Zstandard’s 4,096, because high-bit-depth residuals genuinely require a large alphabet. It also **sizes its tables to the data**: an 8-bit image stored in 16-bit containers (common for MR) shrinks the working tables from ~ 512 KB down to about 2 KB, which keeps everything in fast cache.

Picking the table size automatically. FSE has a “tableLog” knob that sets how large and precise the probability table is. MIC selects it based on how many distinct values it sees and how much data backs each value: it uses a larger table only when there are many active symbols *and* enough data to estimate their probabilities reliably (for example, mammography images after prediction). On the test set this adaptive choice correctly turned on the larger table for exactly the nine images that benefit (1–10% better ratio), and decode speed is unaffected by the choice. Table IV shows the ablation across selected images.

TABLE IV
TABLELOG ABLATION: FORCED $\text{TL}=11/12/13$ VS. ADAPTIVE SELECTION (SELECTED IMAGES, ARM64 REFERENCE PLATFORM).

Image	TL=11	TL=12	TL=13	Adaptive
CR	3.47×	3.63×	3.69×	3.69×
MG1	8.00×	8.57×	8.79×	8.79×
MG2	7.98×	8.55×	8.77×	8.77×
MR2	3.14×	3.24×	3.28×	3.28×
RG2	3.85×	4.12×	4.23×	4.23×
RG3	5.64×	5.95×	6.08×	6.08×

IV. FASTER DECODING: MULTI-STATE DECODING

A. The bottleneck

A standard single-state ANS decoder has an inherent limitation: each symbol depends on the one before it. To decode symbol N the decoder needs the state produced by symbol $N - 1$, so the work cannot start early. With a table lookup taking roughly 4–5 CPU cycles, the loop is limited to about one symbol every few cycles, even though modern CPUs have 4–6 execution ports that remain largely idle. A single decode chain uses only one of them, leaving ~ 75 –83% of the CPU’s capacity unused.

B. The approach

MIC runs **several independent decoders at once** on interleaved positions. With 8 chains, chain 0 handles positions 0, 8, 16, $\dots$; chain 1 handles 1, 9, 17, $\dots$; and so on. The chains share nothing during decoding, each has its own state and its own stream of lookups, so the CPU’s out-of-order engine can keep all of them in flight simultaneously.

On the encoder side, the symbols are split into N interleaved groups, each compressed independently, and the resulting streams are concatenated. They all share one decode table, and each chain’s starting state is recorded in the header. The decoder just writes each chain’s output back to its positions.

C. Measured speedup

In theory, N chains could be up to $N\times$ faster. In practice the speedup is smaller because the chains share some bookkeeping (refilling the bit reader), the unrolled loop puts pressure on the instruction cache, and reading the compressed data has its own bandwidth limit [17]. Measured on the FSE stage alone, **4 states** give +46% on ARM64 and +8% on AMD64 (relative to single state), and **8 states** a further +7% on ARM64 and +2% on AMD64, with **no change to the compressed file**. ARM64 continues to improve as chains double; AMD64 flattens out earlier because its core already extracts substantial parallelism from a single chain. Table V shows the simple latency model against the measured ARM64 speedup.

The decoder signals its mode with a short magic header (a reserved byte value that cannot be a valid single-state setting), so single-, two-, four-, and eight-state files all coexist without confusion. On AMD64 and ARM64 the inner loop uses each architecture’s native variable-shift instructions; other platforms get a pure-Go fallback. Table VI summarises the on-disk layout of a single-frame stream.

TABLE V
MULTI-STATE DECODER LATENCY MODEL AND EMPIRICAL SPEEDUP.

States (k)	Amortized latency	Theoretical	Empirical (ARM64)
1	L cy/sym	$1.0\times$	$1.0\times$
2	$L/2$ cy/sym	$2.0\times$	$1.3\times$
4	$L/4$ cy/sym	$4.0\times$	$1.5\times$
8	$L/8$ cy/sym	$8.0\times$	$1.6\times$

TABLE VI
SINGLE-FRAME MIC BITSTREAM LAYOUT. ALL MULTI-BYTE FIELDS ARE LITTLE-ENDIAN. STATES k IS THE NUMBER OF INTERLEAVED FSE CHAINS ($k \in \{1, 2, 4, 8\}$).

Field	Size	Meaning
Magic / mode byte	1 B	Single-state: encodes <code>tableLog</code> (value 5–17). Multi-state: <code>0xFF</code> .
Multi-state tag	1 B	For multi-state streams, one of <code>0x02</code> , <code>0x04</code> , <code>0x84</code> indicating $k = 2, 4$, or 8 .
Symbol count	4 B	Number of decoded RLE symbols.
TableLog	1 B	Decode-table size: 2^{tableLog} entries (multi-state path only; the single-state path overloads byte 0).
Normalized counts	variable	FSE normalised symbol frequencies, serialised with the standard variable-length encoding.
Per-chain initial states	$k \times 2$ B	Starting state for each FSE chain.
Per-chain stream lengths	$k \times 4$ B	Compressed-stream length per chain (omitted when $k = 1$).
Per-chain compressed streams	variable	Concatenated chain payloads.

V. EVALUATION METHODOLOGY

Dataset. 21 de-identified DICOM images across nine modalities (CT, MR, CR, X-ray, mammography, tomosynthesis, and others), from the public NEMA WG-04 sample sets [22] and a public breast-tomosynthesis case [23]. It is a varied set but small relative to a real clinical archive, so the results are reported as “on this 21-image dataset” rather than as general claims. Table VII lists the images.

TABLE VII
TEST DATASET: 21 CLINICAL DICOM IMAGES SPANNING NINE MODALITIES.

Image	Modality	Dimensions	Raw (MB)
MR	Brain MRI	256×256	0.13
CT	CT scan	512×512	0.50
CR	Comp. radiography	2140×1760	7.18
XR	X-ray	2048×2577	10.1
MG1	Mammography	2457×1996	9.35
MG2	Mammography	2457×1996	9.35
MG3	Mammography	4774×3064	27.3
MG4	Mammography	4096×3328	26.0
CT1	CT scan	512×512	0.50
CT2	CT scan	512×512	0.50
MG-N	Mammography	3064×4664	27.3
MR1	Brain MRI	512×512	0.50
MR2	Brain MRI	1024×1024	2.00
MR3	Brain MRI	512×512	0.50
MR4	Brain MRI	512×512	0.50
NM1	Nuclear med.	256×1024	0.50
RG1	Radiography	1841×1955	6.86
RG2	Radiography	1760×2140	7.18
RG3	Radiography	1760×1760	5.91
SC1	Sec. capture	2048×2487	9.71
XA1	Fluoroscopy	1024×1024	2.00

Baselines. HTJ2K (via OpenJPH [20]), JPEG-LS (via CharLS [21]), and Delta + Zstandard [11] at its maximum level.

All were called in-process (no file or subprocess overhead) for a fair speed comparison.

Metrics. Compression ratio (uncompressed ÷ compressed), and decode and encode throughput in MB/s of uncompressed pixels.

Platforms. Two reference machines: an ARM64 (Apple M4 Pro) and an AMD64 (AWS Intel Xeon 6). Go code used the standard compiler; C components were built with `-O3`. Each benchmark ran the full image 10 times; run-to-run variance stayed under 2% on ARM64 and 5% on AMD64.

The “MIC” column in the result tables is the fastest available C decoder on each platform: the eight-state decoder, with AVX-512 acceleration of the run-length and prediction-reversal stages on AMD64. Both read identical compressed bytes; only the vector width differs.

Browser decoder. A 20 KB pure-JavaScript ES module (zero npm dependencies) plus a 2.5 MB Go WebAssembly build. The JS decoder auto-detects 1/2/4/8-state streams and was benchmarked under Node.js on the ARM64 machine.

VI. RESULTS

A. Compression ratio

JPEG-LS produces the smallest files on every image; its context-adaptive predictor is difficult to beat on pure ratio. MIC, however, is not optimizing ratio alone; it targets the best balance of ratio, speed, and simplicity. Across the dataset MIC averages a **3.46×** ratio: about 91% of JPEG-LS (3.82×) and essentially tied with HTJ2K (3.45×), while decoding faster than both. Mammography compresses best (up to 8.79×) because its smooth tissue produces long near-zero runs. Table VIII gives the per-image ratios for all variants.

B. Encoding speed

MIC encodes faster than the others on most images, on both platforms, because its encoder is a single pass (predict, run-length, table-driven entropy code) while HTJ2K and JPEG-LS do more work per pixel. On **ARM64**, MIC is fastest on all 21 images: roughly 1.9–3.2× HTJ2K, 3–7× JPEG-LS, and 35–240× Delta + Zstandard (whose max-level LZ77 search accounts for its low throughput). On **AMD64**, MIC is fastest on 18 of 21; HTJ2K narrowly leads on the two largest mammography images and one radiography image, because the AMD64 baselines are themselves heavily vectorized. Table IX gives the per-image encoding throughput.

C. Decoding speed

Decoding is where MIC performs best. Its decoder is dominated by the compact FSE loop (one lookup, one bit read, one add) with no hard-to-predict branches. HTJ2K and JPEG-LS both have data-dependent branches that cause CPU mispredictions. Zstandard is the most comparable codec (same predictor, also uses FSE) but works on bytes, so it pays the 2× byte-splitting cost on deep data. On **ARM64**, MIC is fastest on 20 of 21 images (Zstandard is faster only on CT, by 1.3%); roughly 1.5–2.6× HTJ2K, 1.1–1.7× Zstandard, and 2.0–4.5× JPEG-LS. On **AMD64**, MIC is fastest on all 21 images, from

TABLE VIII
LOSSLESS COMPRESSION RATIOS: ALL CODEC VARIANTS, 21 IMAGES. BOLD = BEST RATIO PER ROW.

Image	Raw (MB)	Δ +Zstd-19	MIC	Wavelet	PICS-4	PICS-8	HTJ2K	JPEG-LS
MR	0.13	1.95×	2.35×	2.38×	2.28×	2.21×	2.38×	2.52×
CT	0.50	2.05×	2.24×	1.67×	2.15×	1.96×	1.77×	2.68×
CR	7.18	3.24×	3.69×	3.81×	3.70×	3.71×	3.77×	3.96×
XR	10.1	1.43×	1.74×	1.76×	1.75×	1.76×	1.67×	1.76×
MG1	9.35	7.20×	8.79×	8.67×	8.84×	8.87×	8.25×	8.91×
MG2	9.35	7.21×	8.77×	8.65×	8.83×	8.85×	8.24×	8.90×
MG3	27.3	2.04×	2.24×	2.32×	2.31×	2.34×	2.22×	2.38×
MG4	26.0	3.10×	3.47×	3.59×	3.59×	3.62×	3.51×	3.71×
CT1	0.50	2.47×	2.79×	2.49×	2.54×	2.29×	2.70×	3.19×
CT2	0.50	3.24×	3.49×	2.87×	3.11×	2.72×	3.29×	4.54×
MG-N	27.3	2.04×	2.24×	2.32×	2.31×	2.34×	2.23×	2.38×
MR1	0.50	1.79×	2.09×	2.14×	2.10×	2.08×	2.13×	2.30×
MR2	2.00	2.77×	3.28×	3.34×	3.31×	3.31×	3.35×	3.52×
MR3	0.50	3.47×	3.93×	4.09×	3.89×	3.84×	4.33×	4.51×
MR4	0.50	3.58×	4.12×	4.18×	4.09×	4.03×	4.21×	4.49×
NM1	0.50	4.88×	5.15×	5.02×	5.26×	5.28×	5.76×	6.28×
RG1	6.86	1.45×	1.70×	1.70×	1.70×	1.69×	1.63×	1.72×
RG2	7.18	3.68×	4.23×	4.32×	4.28×	4.30×	4.32×	4.51×
RG3	5.91	5.46×	6.08×	6.82×	6.11×	6.12×	6.99×	7.31×
SC1	9.71	3.46×	3.71×	3.70×	3.73×	3.74×	3.85×	4.73×
XA1	2.00	4.37×	5.01×	4.94×	5.04×	5.03×	4.88×	5.39×
Geomean	—	3.03×	3.46×	3.41×	3.44×	3.38×	3.45×	3.82×

TABLE IX
SINGLE-THREADED ENCODING THROUGHPUT (MB/s). MIC IS THE EIGHT-STATE C ENCODER. BOLD MARKS THE FASTEST CODEC PER ROW ON EACH PLATFORM.

Image	ARM64				AMD64			
	MIC	Δ +Zstd	HTJ2K	JPEG-LS	MIC	Δ +Zstd	HTJ2K	JPEG-LS
MR	504	8	273	97	338	5	229	62
CT	529	8	200	139	292	5	206	97
CR	569	3	204	115	322	2	284	75
XR	690	3	248	125	405	3	250	84
MG1	1,168	14	456	315	546	8	647	209
MG2	1,172	14	454	315	541	8	646	209
MG3	706	3	234	143	332	2	251	104
MG4	993	7	373	215	431	6	415	150
CT1	576	11	238	178	329	7	277	125
CT2	561	8	207	172	286	5	266	125
MG-N	718	2	235	143	339	2	243	104
MR1	677	11	215	120	344	6	244	86
MR2	737	9	231	131	397	4	300	84
MR3	901	22	299	187	428	12	353	123
MR4	711	8	228	175	358	6	318	141
NM1	663	7	247	170	336	5	312	116
RG1	659	5	246	95	392	4	250	65
RG2	811	6	271	151	416	3	356	99
RG3	762	6	285	189	397	4	407	126
SC1	765	6	256	223	407	4	309	168
XA1	693	7	232	152	358	4	310	97
Geomean	740	7	273	161	376	4	311	110

marginally ahead on mammography up to 1.62× on X-ray; switching from four to eight states (plus the AVX-512 widening of the post-entropy stages) is what turned the few remaining close cases into MIC wins.

MIC therefore delivers both **a better ratio than Zstandard on every image** and **faster decoding on nearly every image**, which reflects the 16-bit-native design. JPEG-LS retains about a 10% ratio advantage, so the choice between them comes down to decode speed versus storage cost. Per-image numbers are in Table X, and the entropy stage in isolation is in Table XI.

D. Other MIC variants (for reference)

The main tables use one MIC column, but the codebase includes several variants that exist for different deployment needs rather than to compete on raw speed. The **pure Go**

build runs the entire pipeline in Go with no C dependency, at roughly 40–60% of the C decoder’s speed; this is useful where C cannot be linked (mobile, WebAssembly, locked-down runtimes). The **1/2/4-state decoders** are the same C decoder with fewer chains; they serve as baselines for the multi-state comparison. **Wavelet-based MIC** replaces the predictor with a reversible 5/3 wavelet transform [15] feeding the same back end; it is competitive on a few images but generally slower on this corpus, since smooth medical images do not reward the wavelet’s additional cost. The eight-state C decoder is both the simplest and the fastest configuration on this dataset.

E. Multi-threaded decoding

MIC can also split an image into horizontal strips and decode them on separate threads. This scales nearly linearly up to four

TABLE X

SINGLE-THREADED DECOMPRESSION THROUGHPUT (MB/s). MIC IS THE EIGHT-STATE C DECODER (AVX-512 ACCELERATION OF THE RLE AND DELTA INVERSE STAGES ON AMD64). BOLD MARKS THE FASTEST CODEC PER ROW ON EACH PLATFORM.

Image	ARM64				AMD64			
	MIC	Δ +Zstd	HTJ2K	JPEG-LS	MIC	Δ +Zstd	HTJ2K	JPEG-LS
MR	564	331	301	146	552	357	439	91
CT	466	472	356	187	453	384	329	111
CR	692	564	367	206	566	475	439	123
XR	655	572	366	150	617	508	386	88
MG1	829	691	681	547	913	599	907	332
MG2	873	684	686	557	920	599	896	333
MG3	678	473	362	201	556	358	390	117
MG4	815	581	533	259	676	466	611	157
CT1	571	502	373	238	525	425	390	143
CT2	648	575	372	222	512	456	383	133
MG-N	657	469	360	201	545	355	391	117
MR1	681	437	354	154	603	384	397	94
MR2	747	505	390	224	644	464	491	132
MR3	859	507	453	329	773	485	543	192
MR4	775	523	304	230	632	480	446	153
NM1	866	595	404	277	598	486	467	172
RG1	561	413	354	126	550	347	363	80
RG2	755	574	408	255	730	497	535	154
RG3	740	655	469	320	723	552	628	196
SC1	696	587	384	280	690	484	456	183
XA1	759	611	388	278	688	534	477	165
Geomean	733	557	386	227	602	501	475	144

TABLE XI

FSE DECOMPRESSION THROUGHPUT (MB/s) FOR REPRESENTATIVE MODALITIES, SINGLE-THREAD PURE-Go DECODER. INCREASING THE NUMBER OF INDEPENDENT STATE CHAINS EXPOSES MORE INSTRUCTION-LEVEL PARALLELISM TO THE OUT-OF-ORDER ENGINE.

Image	ARM64 (MB/s)			AMD64 (MB/s)		
	1-st	4-st	8-st	1-st	4-st	8-st
MR	612	953	765	636	899	759
CT	776	1,872	1,729	974	986	953
CR	3,051	5,749	5,521	1,426	1,581	1,638
XR	3,657	6,072	5,807	1,939	1,890	2,248
MG1	3,604	4,557	5,887	2,075	1,521	1,753
MG-N	3,588	6,378	6,307	1,714	2,294	2,473
NM1	1,798	2,535	2,740	1,091	1,274	1,522
RG1	1,786	3,492	5,412	1,198	1,717	2,104
Geomean (21)	2,349	3,422	3,667	1,464	1,575	1,614

threads on both platforms and keeps scaling to eight threads on images above ~ 5 MB, hitting **4–4.6 GB/s** on the largest mammography images. Small images do not benefit: below about half a megabyte, thread setup costs more than it saves. We report this separately because the other codecs run single-threaded in our pipeline, so a direct comparison would not be fair. Table XII gives the per-image strip-parallel numbers.

Practical guidance: to decode many small images, use single-threaded MIC per request, which is already fast. To decode one large image on a multi-core machine, use strip-parallel mode with four to eight threads.

F. In the browser

The JavaScript decoder performs well. Notably, the **four-state** variant is 7–13% *faster* than eight-state in JavaScript, even on the same hardware where eight states win in C. The reason is the JavaScript engine (V8): it can keep four independent chains busy but not eight, so the extra chains add overhead without shortening the critical path. The eight-state path still

works in JavaScript (it lets one decoder accept C-encoded files without conversion), but four states is the best configuration for the browser.

Concretely, a 9.35 MB mammography image decodes in 19.8 ms (473 MB/s) using eight workers with four-state strips. That means a web DICOM viewer can pull a `.mic` file straight from object storage and decode it client-side, with no server doing compression or format conversion. (These numbers are from Node.js; full in-browser benchmarking is future work.) Tables XIII and XIV give the JS numbers.

VII. DISCUSSION

A. Reading the results together

Three observations tie the numbers together. First, MIC’s **ratio advantage over Zstandard** (5–22%, 14% average) comes mainly from working on 16-bit residuals directly instead of splitting them into bytes, and that same choice removes the per-byte branches that slow decoding. Second, on the largest mammography images on AMD64, **HTJ2K’s block coder** regains the speed lead by amortizing its setup over many pixels; MIC stays within 0–20% there. Third, **multi-state decoding** is what makes MIC’s decoder competitive with HTJ2K’s vectorized block coder, and it costs nothing in file size.

B. Choosing a codec

For **low-latency viewing** (PACS, thumbnails, tomosynthesis playback), single-threaded MIC is a sensible default (~ 600 – 900 MB/s per request), with strip-parallel MIC for one large image at a time. For **smallest archival files**, JPEG-LS keeps a $\sim 10\%$ better ratio, at the cost of decoding 1.6 – $5.7\times$ slower on ARM64. Where **DICOM transfer-syntax compatibility** is required, HTJ2K is the appropriate choice, since it is a standard and is particularly strong on large mammography

TABLE XII

MULTI-THREADED DECOMPRESSION THROUGHPUT (MB/s) FOR MIC’S STRIP-PARALLEL MODE. THE “MIC” COLUMN REPRODUCES THE SINGLE-THREADED BASELINE FROM TABLE X EXACTLY; N -STRIP COLUMNS DECODE THE SAME COMPRESSED BYTES IN N PARALLEL THREADS. THE MR IMAGE IS TOO SMALL FOR 8-WAY PARALLELISM ON BOTH PLATFORMS ([†]).

Image	ARM64				AMD64			
	MIC	2-strip	4-strip	8-strip	MIC	2-strip	4-strip	8-strip
MR	564	663	894	702 [†]	552	329	367	294 [†]
CT	466	624	1,157	1,446	453	398	633	714
CR	692	1,052	1,976	3,605	566	758	1,320	2,169
XR	655	1,087	2,024	3,794	617	793	1,515	2,579
MG1	829	1,529	2,529	4,443	913	1,317	1,951	2,890
MG2	873	1,417	2,506	4,585	920	1,356	1,971	3,026
MG3	678	1,112	1,911	3,820	556	838	1,449	2,767
MG4	815	1,326	2,345	4,348	676	1,207	1,897	3,126
CT1	571	819	1,247	1,525	525	458	703	725
CT2	648	864	1,260	1,473	512	481	734	719
MG-N	657	1,147	2,027	3,930	545	913	1,537	2,819
MR1	681	977	1,460	1,851	603	504	761	764
MR2	747	1,137	1,911	3,103	644	778	1,171	1,534
MR3	859	1,205	1,693	2,089	773	591	788	835
MR4	775	997	1,469	1,757	632	540	796	804
NM1	866	1,147	1,682	2,133	598	523	824	762
RG1	561	677	1,203	2,150	550	603	1,101	1,890
RG2	755	1,229	2,156	3,816	730	916	1,473	2,268
RG3	740	1,335	2,343	4,168	723	971	1,618	2,730
SC1	696	1,311	2,206	3,971	690	1,024	1,673	2,541
XA1	759	1,214	1,881	3,191	688	792	1,246	1,748

TABLE XIII

JAVASCRIPT DECODER THROUGHPUT ON THE ARM64 REFERENCE (APPLE M4 PRO, NODE.JS v22.16), FOR THE FOUR-STATE AND EIGHT-STATE FSE VARIANTS. MEDIAN OVER 20 ITERATIONS AFTER THREE WARM-UP RUNS.

Image	Dimensions	Ratio	4-state		8-state	
			ms	MB/s	ms	MB/s
MR	256×256	2.35×	2.5	50	3.3	38
CT	512×512	2.24×	11.9	42	14.1	36
CR	2140×1760	3.69×	136.4	53	151.4	47
MG1	2457×1996	8.79×	68.1	137	78.3	120
MG3	4774×3064	2.29×	514.5	54	563.7	49

TABLE XIV

PICS PARALLEL JAVASCRIPT DECODER (WORKER_THREADS) ON THE ARM64 REFERENCE, WITH FOUR-STATE AND EIGHT-STATE FSE PER STRIP. WORKER COUNT WAS SWEEPED OVER {1, 2, 4, 8, 14}; THE BEST WALL-TIME FOR EACH VARIANT IS REPORTED.

Image	Strips	4-state strips		8-state strips	
		ms	MB/s	ms	MB/s
MR	4	1.1	111	1.3	97
CT	4	4.2	120	4.7	108
CR	8	33.4	215	31.8	226
MG1	8	19.8	473	20.5	457

images on AMD64. Where **no native code is permitted** (browsers, sandboxes), the pure-Go reference or the 20 KB JavaScript decoder runs at about half the C decoder’s speed.

A more structured comparison—a Pareto view of ratio vs. decode speed [24], [25], plus a weighted decision matrix over ratio, speed, platform coverage, and deployment ease [26], [27], following deployment taxonomies for DICOM compression [18], [28]—puts MIC and JPEG-LS on the efficiency frontier (MIC for throughput, JPEG-LS for ratio), with HTJ2K and Zstandard dominated on those two axes. HTJ2K remains the choice whenever standard compatibility is a hard requirement

that the scoring does not capture. Table XV reports the weighted matrix.

TABLE XV

CODEC SELECTION MATRIX. EACH CRITERION IS NORMALISED SO THE BEST CODEC ON THAT ROW SCORES 1.00. QUANTITATIVE ROWS ARE GEOMETRIC MEANS ACROSS THE 21-IMAGE SET. BROWSER COMPATIBILITY: SHIPPED PURPOSE-BUILT JS OR WASM DECODER USED IN PRODUCTION = 1.00; WASM PORT OF THE NATIVE LIBRARY USED IN CLINICAL VIEWERS = 0.85; NO BROWSER PATH = 0.0. DEPLOYMENT EASE: ZERO EXTERNAL RUNTIME DEPENDENCIES = 1.00; UBIQUITOUS SYSTEM DEPENDENCY = 0.90; C/C++ LIBRARY REQUIRING CMAKE BUILD AND LINK INTEGRATION = 0.70. WEIGHT COLUMNS w_P , w_A , w_B REFLECT TYPICAL PRIORITIES FOR (P) PACS VIEWER / LOW-LATENCY DECODE, (A) ARCHIVAL STORAGE, (B) BROWSER DELIVERY.

Criterion	w_P	w_A	w_B	MIC	HTJ2K	JPEG-LS
Compression ratio	0.20	0.50	0.10	0.91	0.90	1.00
Decompression throughput (ARM64)	0.30	0.10	0.15	1.00	0.53	0.31
Encoding throughput (ARM64)	0.05	0.10	0.05	1.00	0.37	0.22
AMD64 native	0.10	0.05	0.05	1.00	1.00	1.00
ARM64 native	0.10	0.05	0.05	1.00	1.00	1.00
Browser compatibility	0.05	0.00	0.40	1.00	0.85	0.85
Deployment ease	0.20	0.20	0.20	1.00	0.70	0.70
Weighted score (P) viewer	—	—	—	0.98	0.74	0.69
Weighted score (A) archival	—	—	—	0.96	0.78	0.80
Weighted score (B) browser	—	—	—	0.99	0.78	0.74

C. Limitations

Small dataset. 21 images across nine modalities is varied but far from a clinical archive; broader validation is needed. **Tuning on the test set.** The adaptive table-size thresholds and the predictor choice were informed by this dataset; the rules are simple (two thresholds, one predictor), which limits overfitting risk, but larger studies should confirm they generalize. **Baseline coverage.** Stronger byte-oriented baselines (Delta + byte/bit-shuffle + Zstd, DICOM RLE) would more sharply isolate the value of the 16-bit entropy stage; left to future work. **Reporting.** Tables give single representative throughput values rather than full confidence intervals. **Not yet a DICOM transfer syntax.**

MIC works on the extracted pixel array; real PACS integration would need a private encapsulation or a standardization step.

VIII. CONCLUSION

MIC is a 16-bit-native lossless codec for medical images: spatial prediction, 16-bit run-length coding, a large-alphabet table-based entropy coder, and a multi-state decoder for speed. On 21 DICOM images it reaches an average ratio of $3.46\times$ (tied with HTJ2K, $\sim 91\%$ of JPEG-LS) while being the fastest decoder on 20/21 images on ARM64 and all 21 on AMD64, and the fastest encoder on all 21 ARM64 images and 18/21 on AMD64. A 20 KB JavaScript decoder and a WebAssembly build show the format can be decoded client-side in a browser. Future work: larger multi-hospital datasets; stronger shuffle-based baselines; full statistical reporting; an ARM64 vector kernel for the wavelet variant; and benchmarking the JavaScript decoder in real browsers. MIC is open source: <https://github.com/pappuks/medical-image-codec>.

REFERENCES

- [1] NEMA, “Digital Imaging and Communications in Medicine (DICOM),” PS3.5—Data Structures and Encoding, <https://www.dicomstandard.org/>, 2024.
- [2] D. S. Taubman and M. W. Marcellin, *JPEG2000: Image Compression Fundamentals, Standards and Practice*. Springer, 2002.
- [3] D. S. Taubman, “High throughput JPEG 2000 (HTJ2K): Algorithm, performance evaluation, and potential,” *Proc. SPIE*, vol. 11137, 2019.
- [4] D. Taubman, A. Naman, M. Smith, P.-A. Lemieux, H. Saadat, O. Watanabe, and R. Mathew, “High Throughput JPEG 2000 for Video Content Production and Delivery Over IP Networks,” *Front. Signal Process.*, vol. 2, Article 885644, Apr. 2022.
- [5] M. J. Weinberger, G. Seroussi, and G. Sapiro, “The LOCO-I lossless image compression algorithm: Principles and standardization into JPEG-LS,” *IEEE Trans. Image Process.*, vol. 9, no. 8, pp. 1309–1324, 2000.
- [6] X. Wu and N. Memon, “Context-based, adaptive, lossless image coding,” *IEEE Trans. Commun.*, vol. 45, no. 4, pp. 437–444, 1997.
- [7] J. Sneyers and P. Wuille, “FLIF: Free Lossless Image Format based on MANIAC Compression,” *Proc. IEEE ICIP*, 2016.
- [8] J. Duda, “Asymmetric numeral systems,” arXiv:0902.0271, 2009.
- [9] J. Duda, “Asymmetric numeral systems: entropy coding combining speed of Huffman coding with compression rate of arithmetic coding,” arXiv:1311.2540, 2013.
- [10] Y. Collet, “Finite State Entropy—A new breed of entropy coder,” <https://github.com/Cyan4973/FiniteStateEntropy>, 2013.
- [11] Y. Collet and M. Kucherawy, “Zstandard Compression and the ‘application/zstd’ Media Type,” RFC 8878, IETF, Feb. 2021.
- [12] F. Giesen, “Interleaved entropy coders,” arXiv:1402.3392, 2014.
- [13] F. Lin, K. Arunruangsilert, H. Sun, and J. Katto, “Recoil: Parallel rANS Decoding with Decoder-Adaptive Scalability,” *Proc. 52nd ICPP ’23*, pp. 31–40, 2023.
- [14] A. Weissenberger and B. Schmidt, “Massively Parallel ANS Decoding on GPUs,” *Proc. 48th ICPP ’19*, Article 100, 2019.
- [15] D. Le Gall and A. Tabatabai, “Sub-band coding of digital images using symmetric short kernel filters and arithmetic coding techniques,” *Proc. IEEE ICASSP*, pp. 761–764, 1988.
- [16] W3C, “Portable Network Graphics (PNG) Specification (Second Edition),” ISO/IEC 15948:2003, Nov. 2003.
- [17] S. Williams, A. Waterman, and D. Patterson, “Roofline: An insightful visual performance model for multicore architectures,” *Commun. ACM*, vol. 52, no. 4, pp. 65–76, Apr. 2009.
- [18] F. Liu *et al.*, “The Current Role of Image Compression Standards in Medical Imaging,” *Information*, vol. 8, no. 4, p. 131, Oct. 2017.
- [19] D. A. Clunie, “Lossless Compression of Grayscale Medical Images: Effectiveness of Traditional and State-of-the-Art Approaches,” *Proc. SPIE Medical Imaging*, 2000.
- [20] A. N. Aous, “OpenJPH: An open-source implementation of High-Throughput JPEG 2000,” <https://github.com/aous72/OpenJPH>, 2024.
- [21] Team CharLS, “CharLS: A C/C++ JPEG-LS library implementation,” <https://github.com/team-charls/charls>, 2024.
- [22] NEMA Working Group 4 (WG-04), “DICOM Compression Sample Images,” National Electrical Manufacturers Association, <https://www.dicomstandard.org/Resources/Open-Content/Compression-Samples>, 2024.
- [23] D. A. Clunie, “Breast Tomosynthesis Case 1,” DICOM sample images, <http://www.dclunie.com/pixelmedimagecases/tomo/case1/>, 2009.
- [24] R. Geldreich, “Quick survey of the lossless decompression Pareto frontier,” 2015, <http://richg42.blogspot.com/2015/11/the-lossless-decompression-pareto.html>.
- [25] J. Sneyers, “JPEG XL and the Pareto front,” Cloudinary Blog, 2022, <https://cloudinary.com/blog/jpeg-xl-and-the-pareto-front>.
- [26] T. L. Saaty, *The Analytic Hierarchy Process: Planning, Priority Setting, Resource Allocation*. New York, NY, USA: McGraw-Hill, 1980.
- [27] K. Sayood, *Introduction to Data Compression*, 5th ed. Burlington, MA, USA: Morgan Kaufmann, 2017.
- [28] D. A. Clunie, “DICOM support for compression: More than JPEG,” *Medical Imaging Informatics and Teleradiology (MIIT)*, 2009, https://www.dclunie.com/papers/MIIT_2009_DicomCompression_Clunie.pdf.